

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Designed and executed **data pipelines and ETL workflows** using **Python** to process, validate, and characterise large datasets from production systems; experience translates directly to materials characterisation data management and experimental result analysis in laser fabrication workflows.
- Developed **automated monitoring and reliability engineering solutions** using **Splunk, CloudWatch, and Dynatrace** for enterprise applications, ensuring **99.5%+ SLA compliance**; this operational excellence mindset applies to ensuring reproducibility and process control in laser-induced graphene fabrication.
- Built and optimised **REST APIs and microservices** using **Java Spring Boot**, leveraging **AWS infrastructure (EC2, S3, Lambda)** and **Docker containerisation** for scalable systems; experience with reproducible, containerised environments is relevant to standardising fabrication methodologies.
- Performed extensive **SQL data analysis** across **PostgreSQL, MySQL, and Oracle** databases, including schema design and ad-hoc query optimisation; skillset supports experimental metadata management and statistical analysis of material properties and sensor performance.
- Managed **CI/CD pipelines using Jenkins** and version control with **Git**, conducting code reviews and ensuring quality gates; this disciplined approach to process control and documentation mirrors best practices in experimental design and material batch tracking.
- Completed **MSc in Data Analytics** with coursework in statistical modelling, machine learning, and data visualisation; combined with **Python (pandas, NumPy)** proficiency, this foundation supports advanced analysis of sensor characterisation, material properties, and e-textile performance metrics.

SKILLS

Programming & Scripting – Python, Java, SQL, shell scripting, Linux.

Data Engineering & Analysis – ETL pipelines, pandas, NumPy, data validation, statistical analysis.

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git.

Databases – PostgreSQL, MySQL, Oracle, schema design, query optimisation.

Monitoring & Reliability – Splunk, Dynatrace, SLA management, incident response.

Frameworks & APIs – Spring Boot, REST APIs, Flask, FastAPI.

Qualifications – MSc Data Analytics (National College of Ireland), BE Computer Science, IELTS Band 7.

EDUCATION AND TRAINING

**MASTER OF SCIENCE in Data Analytics – National College of
Ireland, Dublin**

02/2026

**BACHELOR OF ENGINEERING in Computer Science – SNS College
of Technology, Coimbatore, India**

01/2021

PROJECTS

Enterprise Application Monitoring & Reliability Engineering (HCL Technologies)

- Architected and deployed **Splunk and Dynatrace monitoring solutions** for mission-critical Java/.NET applications in production, achieving **real-time alerting and <99% false-positive rate**.
- Designed **automated incident response workflows** using **Python scripting** that reduced P1/P2 incident response time by **~35%** through intelligent log correlation and metric-based root cause detection.
- Established data-driven SLA dashboards and developed **SQL-based reporting** to track application health, resource utilisation, and performance trends, enabling proactive capacity planning.

Data Pipeline & ETL Automation Framework (HCL Technologies)

- Engineered end-to-end **Python ETL pipelines** to ingest, validate, and transform multi-source operational data; implemented **pandas and NumPy** for efficient data processing across **PostgreSQL and Oracle databases**.
- Developed **scheduled reporting automation** using **Cron jobs and Lambda functions** on **AWS**, reducing manual data extraction effort by **~60%** and ensuring consistency in downstream analytics.
- Built data quality frameworks using **SQL validation rules and Python assertions** to flag anomalies in sensor readings and system metrics, ensuring reliable inputs for analytical and machine learning workflows.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)